Assignment 02

GAMES DEVELOPMENT

SE4031

PHANTOM DASH 2D GAME

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Summary of the Game Concept

Phantom Dash is a 2D platformer game set in a dynamic forest environment where the player must navigate through multiple levels filled with obstacles, traps, and enemies to reach the final objective. The gameplay focuses on smooth character movement, increasing difficulty, and interactive feedback systems.

The player must survive while maintaining a high score, avoiding damage, and progressing through different rooms using a door-based level system. The game combines traditional platforming mechanics with modern features such as dynamic difficulty, visual feedback, and scoring systems to enhance engagement.

The ultimate goal of the player is to reach the final level and collect a special object (such as a magical artifact) while achieving the highest possible score.

Description of Core Features Implementation

1. Player Movement System

The player movement system is implemented using Unity's Rigidbody2D physics system. The system provides smooth and responsive control over the character.

Features include:

- Horizontal movement using keyboard input
- Jumping with adjustable jump height
- Coyote time, allowing the player to jump shortly after leaving a platform
- Multiple jumps (double jump)
- Wall detection and wall jumping mechanics

These features are implemented in the PlayerMovement.cs script, ensuring fluid and responsive gameplay.

2. Health System

The health system is implemented in Health.cs and manages the player's survival.

Key functionalities:

- Player loses health when hit by enemies or traps
- Temporary invulnerability (iFrames) after taking damage
- Visual feedback (flashing effect) when damaged
- Death animation when health reaches zero

The system ensures that the player receives clear feedback when taking damage.

3. Respawn System

The respawn system allows the player to continue gameplay after dying. Implemented in PlayerRespawn.cs:

- Player respawns at the last activated checkpoint
- Only **one respawn** is allowed per game session
- If the player dies again after respawning → Game Over

This adds a layer of challenge and strategic gameplay.

4. Score System

The score system is implemented using ScoreManager.cs.

Features:

- Score starts at **100**
- Score decreases when the player takes damage
- Score is displayed in real-time during gameplay
- Final score is displayed on the win screen
- If score reaches **0**, the game ends immediately

This system encourages careful gameplay and rewards skillful performance.

5. Multiplier System

The multiplier system enhances scoring by rewarding active gameplay.

Key behavior:

- Multiplier increases over time **only when the player is moving**
- Prevents idle score farming
- Multiplier resets when the player takes damage
- Bonus score is added based on multiplier level

This feature improves engagement and introduces a skill-based scoring mechanism.

6. Level Transition System

The level transition system is handled using doors between rooms. Implemented in Door.cs:

- Player enters a door trigger to move between rooms

- Camera smoothly transitions to the next room
- Only the active room remains enabled

This creates a structured level progression system.

7. Game Over System

Implemented in UIManager.cs. Game Over is triggered when:

- Player dies twice
- Score reaches 0

Features:

- Displays Game Over screen
- Allows player to restart or return to main menu

8. Win Condition

The win condition is triggered when the player reaches the final object.

Features:

- Displays a **Congratulations screen**
- Shows the final score
- Pauses the game

This provides a satisfying end to gameplay.

Control Guide

Actions	**Keys**
Move Up	W / Up Arrow
Move Down	S / Down Arrow
Select	E
Attack	Left Click
Jump	Space
Pause	Escape

Description of Creative Features

1. Near Death Visual Effect

A dynamic red overlay appears on the screen when the player's health decreases.

- The intensity increases gradually based on remaining health
- Provides real-time visual feedback
- Enhances immersion and tension

2. Score Multiplier System

The multiplier system rewards players for continuous movement.

- Increases over time when the player is moving
- Stops increasing when the player is idle
- Resets when the player takes damage
- Adds bonus score dynamically

This feature promotes active gameplay and strategic movement.

3. Dynamic Player Speed Increase

The player's speed increases automatically over time.

- Speed increases only when the player is moving
- Gradually increases difficulty
- Keeps gameplay engaging and challenging

4. Player Hit Feedback (Shake Effect)

When the player is hit by an enemy or trap:

- The player briefly shakes
- Provides immediate visual feedback
- Enhances game feel and responsiveness

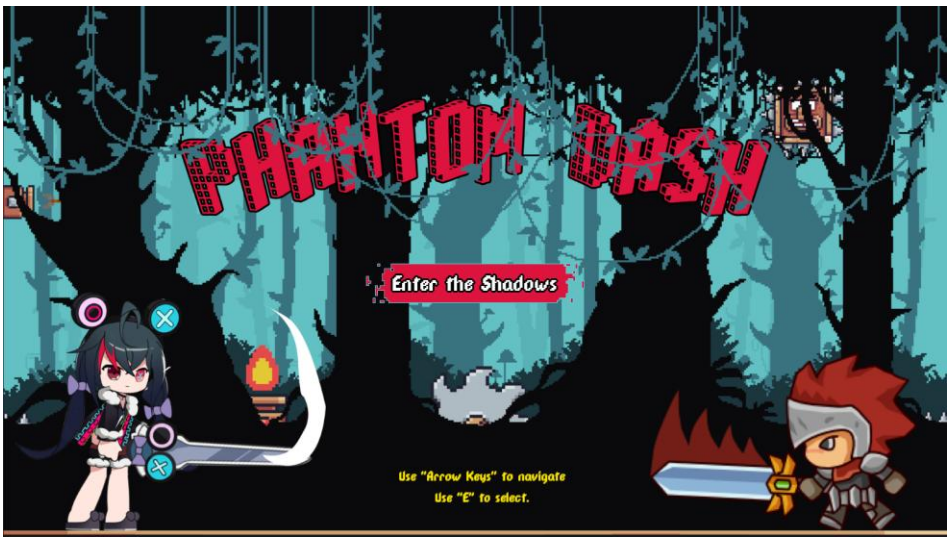
Tools Used

- **Game Engine:** Unity 2022.3 LTS
- **Programming Language:** C#
- **IDE:** Visual Studio

Video Link

Phantom Dash 2D

Screenshots



Multiplier: x1
Score: 10

Multiplier: x1
Score: 0
FATE HAS CLAIMED YOU
Restart
Return to Base
Exit
Use "Arrow Keys" to navigate
Use "E" to select.

PHANTOM DASH
Congratulations !
Final Score:
10
Return to Base
Exit